This PDF combines selected excerpts from different students' Lab 1 reports.

Lab 1 | Selected Report Showcase

Source guide

Steps 1 and 4: Nathan Gabrielle

Step 2: Nathan Gabrielle and Sibo Cao

Steps 3 and 5: Jiang Pinheng

Step 6: Sibo Cao

Step 7: Nathan Gabrielle

Step 8: Jiang Pinheng (Q1-Q2); Zhu Zhexian (Q3)

Step 1 | Collect Dataset 1

Nathan Gabrielle | Original page 1

II. Step 1: Dataset 1 / Model 1

Class	Number of Uploaded Samples
Mandatory	100
Prohibitory	100
Warning	100

Table 1: Dataset 1 class and uploaded samples.

Description:

Dataset 1 is collected through Google. We collected 100 samples for each class, for a total of 300 samples. All these images are flat 2D stock vector illustrations of traffic signs, centered in the middle of the image, with a plain one-color background.

Step 2 | Collect a Fixed External Test Dataset

Selected descriptions and condition records from two reports

Nathan Gabrielle | Original page 1 | Sample counts and data source

III. Step 2: Fixed External Test Dataset

Class	Number of Test Samples
Mandatory	20
Prohibitory	20
Warning	20

Table 2: Fixed External Test Dataset

Description:

This fixed test dataset is collected through the publicly available dataset “CCTSDB”. We selected images from the 5000-6000 range, so they won’t be mixed up with the training dataset for other models. These images are randomly chosen, 20 images/class, totaling 60 images for the test dataset. These images are then cropped (1:1) so that the signs are centered in the middle of the image, more clearly visible, and zoomed in.

Sibo Cao | Original page 2 | Collection description and condition-count table

A separate external test dataset is collected from the cctsdb dataset. These images were not uploaded to Teachable Machine. They were used only in the test of the trained model.

All the images are not duplicated or closely resemble, with images including different angles, lighting conditions, distances, backgrounds, and levels of clarity. They are very close to everyday situations.

Table 3: External Test Collection Conditions

Collection Condition	Mandatory Signs	Prohibitory Signs	Warning Signs
Non-frontal angle	3	3	3
Low light	4	3	5
Far distance	4	3	3
Low Resolution Images	3	6	1
Has Obstacles	2	3	1

Step 3: Train and Evaluate Model 1

Model 1 was trained in Teachable Machine using Dataset 1 with the following Advanced settings: 50 epochs, batch size 16 and learning rate 0.001. Teachable Machine reported an internal test accuracy of 100% on its own 15% split. The trained project was saved as Model1.tm. The model was then evaluated with every sample of the fixed external test dataset using the Preview panel: each of the 60 images was shown once and its predicted class was compared with the true class. Table 4 records the configuration and the results, Table 5 gives the per-class results, and equation (1) shows the external test accuracy calculation as defined in the lab handout.

Table 4: Model 1 – training configuration and evaluation results.

Item	Value
Dataset size (uploaded samples)	300 images (100 per class)
Fixed external test dataset size	60 images (20 per class)
Epochs	50
Batch size	16
Learning rate	0.001
Correct predictions (external test)	23 of 60
Internal test accuracy (Teachable Machine)	100% (45/45 internal test samples; 15/15 per class)
External test accuracy	38.33% (see equation (1))

External test accuracy = (Number of correct predictions / Total number of external test samples) × 100% (1)

Model 1: 23 / 60 × 100% = 38.33%.

Table 5: Model 1 – external test results per class (20 test samples per class).

Class	Correct / total	Accuracy	Misclassified samples
Standing	0 / 20	0%	All samples 1–20
Walking	20 / 20	100%	None
Fallen	3 / 20	15%	Samples 4–20 (only samples 1–3 correct)
Overall	23 / 60	38.33%	—

V. Step 4: Dataset 2 / Model 2

Class	Number of Uploaded Samples
Mandatory	100
Prohibitory	100
Warning	100

Table 4: Dataset 2 class and uploaded samples.

Description:

Dataset 2 is collected through the “CCTSDB” dataset. We selected 100 samples for each class, for a total of 300 samples from the 1000-1999 range. All these images are cropped so that the sign is clearly visible and zoomed in. Approximately 90% of the images have the sign centered, while the rest have it positioned on the sides. These images were taken in a real environment (with mixed sunny and rainy conditions), so they have a real world background.

Step 5: Train and Evaluate Model 2

Model 2 was trained in Teachable Machine using Dataset 2 with exactly the same settings as Model 1 (50 epochs, batch size 16, learning rate 0.001). Teachable Machine reported an internal test accuracy of 95.6% (per-class accuracies: Standing 0.87 = 13/15, Walking 1.00, Fallen 1.00; 43 of 45 internal test samples correct). The trained project was saved as Model2.tm. The same fixed external test dataset of 60 images was then evaluated once with the Preview panel. Table 7 records the configuration and results and Table 8 gives the per-class results.

Table 7: Model 2 – training configuration and evaluation results.

Item	Value
Dataset size (uploaded samples)	300 images (100 per class)
Fixed external test dataset size	60 images (20 per class)
Epochs	50
Batch size	16
Learning rate	0.001
Correct predictions (external test)	49 of 60
Internal test accuracy (Teachable Machine)	95.6% (43/45 internal test samples; Standing 13/15, Walking 15/15, Fallen 15/15)
External test accuracy	81.67% (49 / 60 × 100%)

Table 8: Model 2 – external test results per class (20 test samples per class).

Class	Correct / total	Accuracy	Notes
Standing	10 / 20	50%	Correct for samples 4–6 and 13–19; incorrect for 1–3, 7–12 and 20
Walking	19 / 20	95%	Incorrect only for sample 19
Fallen	20 / 20	100%	All samples correct
Overall	49 / 60	81.67%	—

Step 6: Collect Dataset 3

Table 9: Class definition and Dataset 3 size

Class	Number of uploaded samples	Teachable machine training (85%)	Internal test (15%)
Mandatory Signs	100	85	15
Prohibitory Signs	100	85	15
Warning Signs	100	85	15
Total	300	255	45

Dataset 3 is taken from the 2000 – 2999 range of dataset cctsdb. The difference between dataset 2 and dataset 3 is that dataset 3 is not cropped or zoomed in. The images in dataset 3 are just raw images as provided in the database cctsdb. The conditions are more varied than model 2, including more angle and distance changing, with unpredictable position of the sign.

Table 10: Model 3 Settings and Results

Item	Value
Dataset 3 size	300
Epochs	50
Batch Size	16
Learning Rate	0.001
Internal test accuracy	≈71.33%

Table 11: Model 3 Fixed External Testing Result

Sign type	The number of correct predictions
Mandatory Signs	14
Prohibitory Signs	14
Warning Signs	16
External test accuracy	73.33% (approximately)

External Test Accuracy = 44/60 ×100%≈ 73.33%

The external test accuracy results show that model 3 performs better than model 1 but worse than model 2.

Step 7 | Compare and Analyse the Models

Nathan Gabrielle | Original pages 4-5 | Selected excerpts from source section "Step 8"

IX. Step 8: Compare and Analyze the Models

Model	Dataset Size	External Test Size	Epochs	Batch Size	Learning Rate	Correct Predictions	Internal Test Accuracy	External Test Accuracy
Model 1	300	60	50	16	0,001	40	100%	66,7%
Model 2	300	60	50	16	0,001	51	100%	85%
Model 3	300	60	50	16	0,001	44	100%	73,3%

Table 8: Comparison of all models

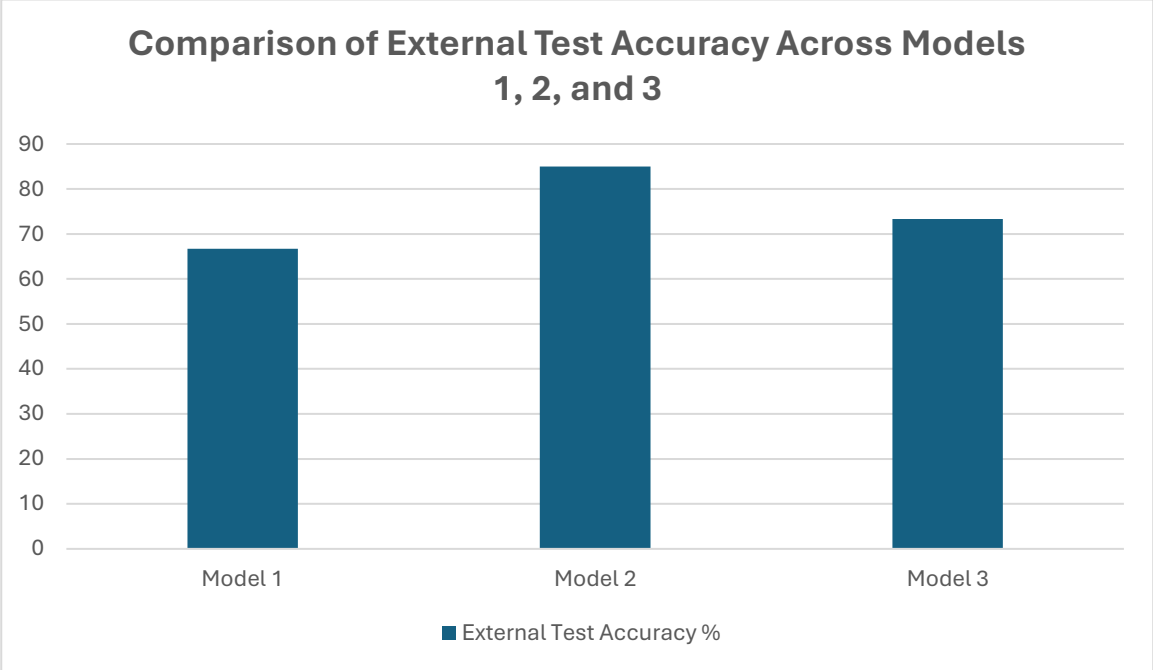


Figure 1: Comparison of external test accuracy (%) across Models 1, 2, and 3

Correctly Classified External Test Samples

Model 2 achieved the highest external test accuracy (85%, 51/60 correct), and its correctly classified samples were those captured under clear, and well lit conditions. The external test dataset (Table 2) was drawn from the CCTSDB dataset and cropped to a 1:1 ratio, with approximately 90% of signs centred and 10% positioned toward the sides. Model 2 correctly classified 19/20 mandatory, 19/20 prohibitory, and 13/20 warning samples.

Incorrectly Classified External Test Samples

Warning signs (yellow triangles) were the most frequently misclassified class across all models. Model 2 misclassified 7/20 warning signs, Model 3 misclassified 6/20, and Model 1 misclassified 12/20.

Model 1 (trained on flat vectors) misclassified the most samples overall. The errors were not only in one class but spread across all three. This is most likely because the model struggled with the domain change from 2D vector graphics to real world photographs.

Step 8 | Reflection Questions

Selected explanations of experimental control, independent testing, and confidence

Q1 | Jiang Pinheng | Original page 13 | Opening explanation

Question 1 — Why must the class definitions, number of uploaded samples, training parameters and fixed external test dataset remain unchanged when comparing Model 1 and Model 2?

Because the comparison should be a controlled experiment in which the only varying factor is the dataset (i.e. the data-collection strategy). If class definitions changed, the two models would solve different tasks; if the number of uploaded samples changed, the comparison would also reflect a change in data volume; if the epochs, batch size or learning rate changed, the comparison would reflect a change in optimisation rather than data; and if the external test dataset changed, the accuracy values would measure performance on test sets of different difficulty and would no longer be comparable.

Q2 | Jiang Pinheng | Original page 13 | Opening explanation

Question 2 — If you repeatedly redesign the uploaded datasets after examining errors on the fixed external test dataset, is the test dataset still completely unseen? Why would an additional untouched test dataset provide stronger evidence of real-world performance?

No. Once the errors on the external test dataset are used to decide which conditions to collect, which samples to add or remove or which images to keep, information from the test dataset has flowed into the training data. The test dataset then behaves like a validation set used for model selection: performance on it becomes optimistically biased, because the dataset has influenced the decisions that produced the model. An additional untouched test dataset, which has never been examined during development, gives a stronger and more honest estimate of real-world performance, because no decision has been adapted to it.

Q3 | Zhu Zhexian | Original page 9 | Complete response

3. Does a higher confidence score always indicate better model performance? Explain the difference between confidence and accuracy, and support your answer with at least one result from Model 1 or Model 2.

Answer: A higher confidence score does not always indicate better model performance. Confidence is an internal prediction score that reflects how certain a trained AI model is about its prediction for a particular test sample, but it does not guarantee that the prediction is correct. Accuracy, by contrast, measures the proportion of correct predictions across the test dataset rather than how confident those predictions are. For example, when Model 2 was evaluated using the external test dataset, its confidence scores were almost always close to 100%, even when its predictions were incorrect.